

Human capacity for explosive force production: Neural and contractile determinants

J. P. Folland¹, M. W. Buckthorpe¹, R. Hannah^{1,2}

¹School of Sport, Exercise and Health Sciences, Loughborough University, Loughborough, UK, ²Sport, Health and Performance Enhancement (SHAPE) Research Group, School of Science and Technology, Nottingham Trent University, Nottingham, UK
Corresponding author: Jonathan Folland, School of Sport, Exercise and Health Sciences, Loughborough University, Ashby Road, Loughborough, Leicestershire LE11 3TU, UK. Tel: +44 (0)1509 226334, Fax: +44 (0)1509 226301, E-mail: j.p.folland@lboro.ac.uk

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This study assessed the integrative neural and contractile determinants of human knee extension explosive force production. Forty untrained participants performed voluntary and involuntary (supramaximally evoked twitches and octets – eight pulses at 300 Hz that elicit the maximum possible rate of force development) explosive isometric contractions of the knee extensors. Explosive force ($F_{0-150\text{ ms}}$) and sequential rate of force development (RFD, 50-ms epochs) were measured. Surface electromyography (EMG) amplitude was recorded (superficial quadriceps and hamstrings, 50-ms epochs) and normalized (quadriceps to M_{max} , hamstrings to EMG_{max}). Maximum voluntary force (MVF) was also assessed. Multiple linear regressions assessed the significant neural

and contractile determinants of absolute and relative (%MVF) explosive force and sequential RFD. Explosive force production exhibited substantial interindividual variability, particularly during the early phase of contraction [F_{50} , 13-fold (absolute); 7.5-fold (relative)]. Multiple regression explained 59–93% (absolute) and 35–60% (relative) of the variance in explosive force production. The primary determinants of explosive force changed during the contraction (F_{0-50} , quadriceps EMG and Twitch F; RFD₅₀₋₁₀₀, Octet RFD₀₋₅₀; $F_{100-150}$, MVF). In conclusion, explosive force production was largely explained by predictor neural and contractile variables, but the specific determinants changed during the phase of contraction.

Explosive force production is considered integral to the performance of tasks where the time available to develop force is limited. This includes athletic events such as jumping and sprinting (Pääsuke et al., 2001; Tillin et al., 2013a), but may be equally important when stabilizing the musculo-skeletal system during standing, locomotion, or in response to mechanical perturbation (Fleming et al., 1991; Izquierdo et al., 1999; Chang et al., 2005; Pijnappels et al., 2008), and thus, the prevention of falls and injury. Because of these wide-ranging implications, a good understanding of the variability in explosive force production between individuals and the precise determinants of explosive force production are sought. To date, the interindividual variability in explosive force production and the importance of neural and contractile characteristics to this variability has not been elucidated.

Explosive force can be defined as the capability to increase contractile force from a low or resting level as quickly as possible. This parameter is typically measured isometrically and has been quantified as the force (or torque) produced at specific time points from contraction onset (Tillin et al., 2010; Hannah et al., 2012), the rate of force development (RFD) over a particular

time period [i.e., the gradient of the force-time trace (Aagaard et al., 2002)], or the impulse/integral beneath the force time recording (Aagaard et al., 2002; de Ruiter et al., 2004). The interindividual variability in explosive force production has not been documented, although our previous work has suggested there may be considerable variability between individuals even in homogenous cohorts (Tillin et al., 2010). Documenting the variability is a prerequisite to understanding the specific determinants of explosive force production, with preliminary evidence indicating that several factors may influence explosive force, including maximum voluntary force [MVF (Andersen & Aagaard, 2006)], morphological factors such as fiber type composition (Viitasalo & Komi, 1978; Harridge et al., 1996) and muscle-tendon unit (MTU) stiffness (Bojsen-Møller et al., 2005), neural drive to the involved musculature (de Ruiter et al., 2007; Klass et al., 2008), and the contractile response to an evoked twitch (Andersen & Aagaard, 2006).

The combined influence of morphological factors can be studied by examining the intrinsic contractile properties of the MTU in response to an evoked contraction. The peak RFD in response to an evoked twitch

contraction has been found to explain 20–36% of the variance of explosive voluntary force during the first 50 ms of contraction, but not at later time points (Andersen & Aagaard, 2006). However, as an evoked twitch generates a relatively low RFD, it may provide only limited information on the capacity of the MTU for explosive force production. High frequencies of stimulation are known to be required for maximal rates of force development to be achieved (de Ruiter et al., 1999; Deutecom et al., 2000), and an evoked octet of eight pulses at 300 Hz has been shown to drive the MTU at its maximum possible RFD (de Ruiter et al., 2004), but how this contractile characteristic relates to, or influences, voluntary explosive force production remains unknown.

Neural drive to the agonist muscles may be important for maximizing explosive force production (Zehr & Sale, 1994; Van Cutsem et al., 1998; Aagaard et al., 2002), and unusually, high-motor unit firing frequencies have been recorded during the initial phase of rapid or explosive contractions (Desmedt & Godaux, 1977; Van Cutsem et al., 1998). Neuromuscular activation of the agonist muscles, assessed with surface electromyography (EMG), has been suggested to explain a moderate to high proportion of the variance in explosive force (de Ruiter et al., 2006, 2007; Klass et al., 2008). However, these studies involved small groups for the assessment of relationships ($n \leq 11$), and only considered explosive force during the early phase of contraction [40 ms (de Ruiter et al., 2006, 2007); <80 ms (Klass et al., 2008)] or did not document the contribution of antagonist muscle co-activation. Therefore, a thorough evaluation of the contribution of neural factors including agonist and antagonist EMG throughout the time course of explosive contraction seems required.

Moreover, no previous studies have addressed the relative importance of neural and contractile characteristics in determining explosive force production. It seems likely that the contribution of neural and intrinsic contractile properties may change throughout the rise of the force-time curve (Andersen & Aagaard, 2006), and therefore, considering the relationship of these characteristics to force measurements at a range of time points from contraction onset may be insightful. The explosive force achieved at a specific point in time likely reflects the combined influence of contributory variables prior to that instant, whereas sequential RFD, assessed over discrete consecutive time periods, may more effectively isolate any changing contribution of neural and contractile determinants. For example, after the early phase of contraction, neural activation may be high/maximal in all individuals, and thus the determinants of later phase RFD may be more dependent upon contractile factors. The sequential RFD during later time periods could also be influenced by the force already achieved prior to that period (Tillin et al., 2010). A more thorough understanding of the specific determinants of explosive force production may facilitate development of interventions

(e.g., exercise training) designed to improve explosive force production.

The aim of this study was to document the interindividual variability in explosive force production of the knee extensors, and assess the integrative neural and contractile determinants of explosive force production expressed in absolute and relative terms. Relative values, corrected for maximum voluntary force, may further highlight the comparative importance of neural and intrinsic contractile characteristics. The influence of neural (agonist quadriceps and antagonist hamstrings EMG) and intrinsic contractile characteristics (evoked Twitch and Octet responses), as well as contractile capacity (MVF) and contractile history (Prior Force), in determining explosive force production and the rise of the force-time curve (sequential RFD) were assessed.

Methods

Participants

Forty participants (20 males and 20 females) volunteered for this study and they were healthy and free from musculo-skeletal injury to the lower limb; 23 ± 3 years of age; 173 ± 9 cm tall; and had a body mass of 67.1 ± 9.8 kg. Participants provided written informed consent prior to their involvement in the study, which complied with the Declaration of Helsinki and was approved by the Ethical Advisory Committee of Loughborough University. In order to control for the potentially confounding effects of training background, we studied untrained individuals with similar, low to moderate levels of habitual physical activity, who had no history of systematic strength or power training. Physical activity was assessed using the International Physical Activity Questionnaire [Short Format, <http://ipaq.ki.se/downloads.htm> (Craig et al., 2003)] that enabled estimation of participants' average energy expenditure as 1868 ± 1025 metabolic equivalents-mins/week. To control for the possibility of menstrual cycle phase influencing the neuromechanical function of the knee extensors (e.g., Phillips et al., 1996; Sarwar et al., 1996), female participants were required to have been taking the combined monophasic oral contraceptive pill for ≥ 6 months, and were tested between days 7–21 of pill consumption (Onambele et al., 2007).

Study design

Participants visited the laboratory for 60–90 min on two separate occasions, 7 days apart, to complete a familiarization session and a test session. Both sessions commenced at a consistent time of the day (12:00–16:30 h) following at least 36 h without strenuous exercise, and involved the same protocol; however, only measurements from the test session were used for analysis. Participants were seated in a static dynamometer and completed a series of voluntary and electrically evoked isometric contractions of the knee extensors of their dominant leg. Specifically, maximum voluntary contractions (MVCs), explosive voluntary contractions, evoked twitch, and evoked octet contractions were performed in this order. The MVCs were used to determine MVF and the explosive contractions were used to determine explosive force during the first 150 ms of contraction, and sequential RFD in consecutive 50-ms periods (0–50, 50–100, and 100–150ms). Separate contractions were utilized because of the influence of instruction and the task goal on the measured outcome (Sahaly et al., 2001). In addition, a series of knee flexion MVCs were performed at the end of each session. Knee extension/flexion force, as well as surface

EMG of the superficial knee extensor and flexor muscles, was recorded throughout all the contractions.

Knee extension and flexion force

Participants were firmly secured in a custom-built isometric dynamometer (Parker et al., 1990), with hip and knee joint angles of 100° and 85° (full extension = 180°), respectively. Adjustable straps across the pelvis and shoulders were tightly fastened to prevent extraneous movement. A strap, 40 mm width reinforced canvas webbing, was placed proximal to the ankle (15% of tibial length above the medial malleolus), positioned perpendicular to the tibia and in series with a calibrated strain gauge (Jones & Parker, 1989). This low-noise strain gauge (range of baseline noise <0.2 N or <0.02% full scale deflection (Tillin et al., 2010) consisted of an U-shaped aluminum beam with two high-frequency response silicon strain gauges bonded either side of the horizontal section. The force signal was amplified ($\times 1000$), sampled at 2000 Hz using an external A/D converter (16-bit signal recording resolution; Micro 1401, CED, Cambridge, UK) and interfaced with a personal computer (PC) using Spike 2 software (CED). The strain gauge was calibrated on a weekly basis by hanging a series of weights of known mass (0–85 kg) off the strain gauge. The strain gauge was linear up to 850 N (mean $R^2 = 1.000$) and very stable [coefficient of variation (CV) of the calibration coefficient <0.5%]. For all knee extension and flexion recordings, the baseline resting force was subtracted from all active force recordings to provide gravity correction. Force measurements made in this way appear to be less influenced by the variability in body size than torque values (i.e., lower power function exponents for force than torque when scaled to body mass or height; Jaric et al., 2002; Folland et al., 2008). Furthermore, normalization of each individual's explosive force values to MVF would be expected to eliminate any confounding influence of body size on explosive force production.

EMG

Electromyographic signals were recorded from the superficial knee extensors [rectus femoris (RF), vastus medialis (VM), vastus lateralis (VL)] and flexors [biceps femoris (BF) and lateral hamstrings (semitendinosus and semimembranosus, LH)] using two Delsys Bagnoli-4 Systems (Delsys, Boston, Massachusetts, USA). After preparation of the skin by shaving, light abrasion and cleaning with alcohol, double-differential surface electrodes (three contacts with a 1 cm inter-electrode distance, Model DE-3.1; Delsys) were attached over the belly of each muscle, parallel to the presumed orientation of the muscle fibers, and at standardized percentages of thigh length measured from the knee joint space to greater trochanter (RF, 60%; VM, 20%; VL, 50%; BF and LH 50%). These sites were selected to avoid the innervation zones of each of the assessed muscles (Rainoldi et al., 2004). The isometric dynamometer employed in this study has a section of the seat base removed in the mid-thigh region to facilitate hamstrings EMG recordings without compression or movement artifacts during seated contractions of the thigh musculature. The reference electrode was placed on the patella of the same limb. The raw EMG signals were amplified ($\times 1000$, differential hardware amplifier, 20–450 Hz) and sampled at 2000 Hz using the same A/D converter and PC software that enabled synchronization with the force recordings. EMG data were band-pass filtered between 6 and 500 Hz using a fourth-order zero-lag Butterworth filter prior to analysis.

Protocol

Measurements were completed in the following order:

Knee extension maximum voluntary contractions

Participants performed a series of warm-up contractions at progressively higher levels of force before completing three MVCs, each lasting >3 s and proceeded by ≥ 30 -s rest. Participants were instructed to push as hard as possible for at least 3 s. Typically, they reach a force plateau in <1 s and received strong verbal encouragement to increase the force further during the subsequent 2 s. Participants also received real-time biofeedback regarding the force response during and after each contraction. MVF was defined as the instantaneous highest force recorded during any of the MVCs or explosive contractions. The root mean square (RMS) of the EMG signal for each muscle (RF, VM, VL) was calculated over a 500-ms epoch surrounding MVF (250 ms either side), normalized to $M_{\max}$ (Folland & Williams, 2007; see below) and averaged for the quadriceps group ($QEMG_{MVF}$).

Knee extension explosive voluntary contractions

Participants completed 10 explosive isometric contractions of the knee extensors each separated by 20-s rest [an example force recording is shown in Fig. 1(a)]. Participants were instructed to extend their knee “as fast and hard as possible” for 1 s upon hearing an auditory signal, with the emphasis on “fast.” In order to provide biofeedback on their explosive performance, the slope of the force-time curve (2 ms time constant) was displayed throughout these contractions, with the peak slope (peak RFD) of their best attempt highlighted with an on-screen cursor. Participants were instructed to avoid any prior countermovement, and the resting force level was displayed on a sensitive scale to provide biofeedback on whether any countermovement or pre-tension had occurred. Finally, a visual marker on the screen depicted 80% of MVF during the contractions, and participants were instructed to achieve a peak force above this level during each explosive contraction. The explosive contractions were performed until 10 contractions meeting these criteria had been completed (up to a maximum of 13 contractions).

The three explosive voluntary contractions with the highest peak RFD were used for analysis, with force and EMG measurements taken at specific point/periods from their respective onsets. Force and EMG measurements were averaged across these three contractions. Signal onsets of all the voluntary and evoked contractions were identified visually (see below). Force was measured at 25, 50, 75, 100, and 150 ms from its onset (defined as F_{25} , F_{50} , F_{75} , etc.). RFD ($\Delta \text{force} / \Delta \text{time}$) was quantified for three sequential 50-ms time periods (RFD_{0-50} , RFD_{50-100} , and $RFD_{100-150}$). Prior Force was defined as the force achieved immediately before subsequent RFD measurements (e.g., F_{100} for $RFD_{100-150}$). Explosive force and sequential RFD measurements were expressed in absolute terms and relative to MVF.

Example EMG recordings from the agonist muscles (VM, VL, and RF) during an explosive voluntary contraction are shown in Fig. 1(b). The RMS of the EMG signal for each muscle (RF, VM, VL, BF, and LH) was calculated for three consecutive 50-ms time periods from EMG onset – defined as the onset of the first muscle to be activated. EMG values from the agonist quadriceps (RF, VM, VL) were expressed as a percentage of their respective maximal M-wave ($M_{\max}$; see below) and antagonist hamstrings (BF and LH) EMG was expressed as a percentage of maximal EMG recorded at knee flexion MVF (detailed below), prior to calculating mean quadriceps ($QEMG$) and hamstrings ($HEMG$) values for each time period. The rate of change in quadriceps and hamstrings EMG ($\Delta QEMG / \Delta t$ and $\Delta HEMG / \Delta t$) between sequential time periods was also calculated.

Electrically evoked twitch and octet contractions

Discrete electrical stimuli were delivered via percutaneous stimulation of the femoral nerve in the femoral triangle to evoke

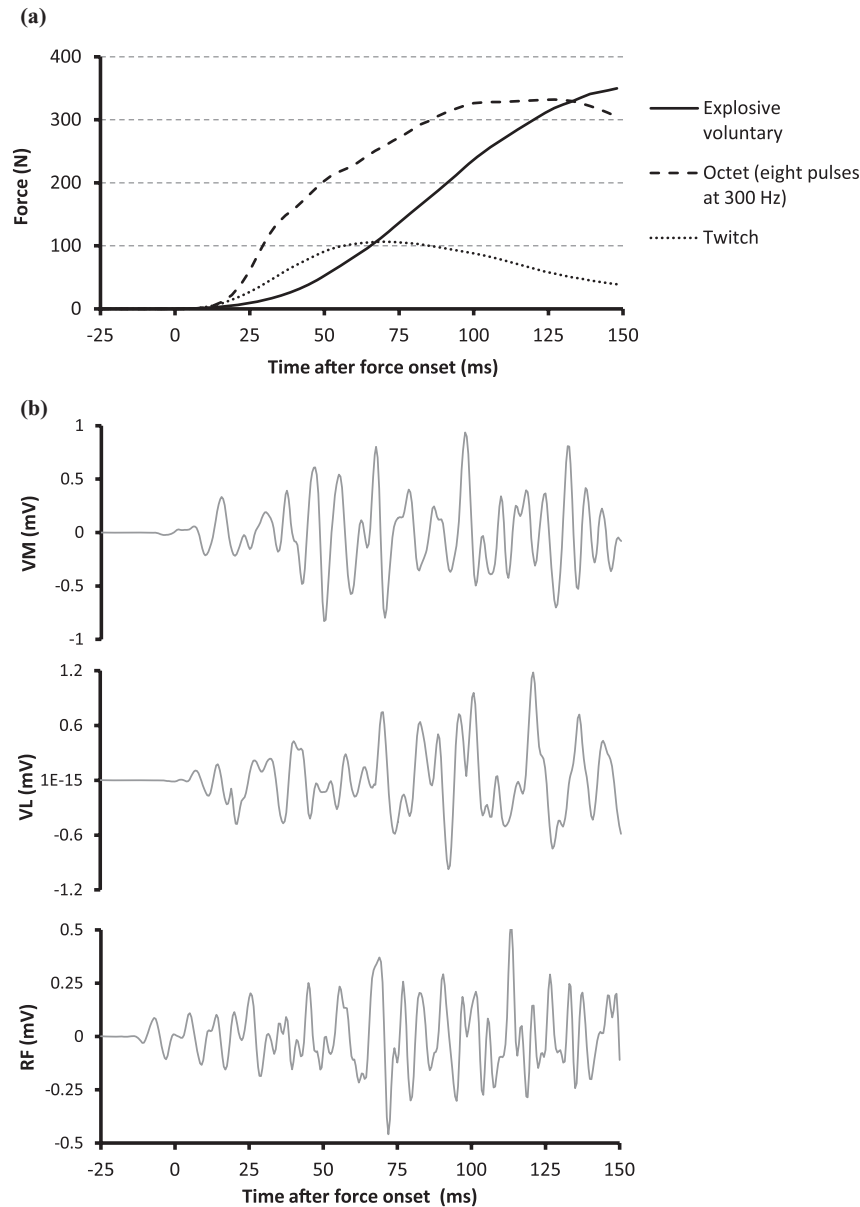


Fig. 1. Example recordings of (a) force during an explosive voluntary contraction and during evoked twitch and octet contractions (measures of the intrinsic contractile properties) and (b) EMG of the agonists (superficial quadriceps: VM, VL and RF) during the first 150 ms of explosive voluntary contraction for one participant. Note the substantially steeper rise in evoked octet force, representing the maximum capacity for force development by the MTU, than explosive voluntary force.

supramaximal twitch and octet contractions of the knee extensors [example force recordings are shown in Fig. 1(a)]. Stimulation was achieved via a cathode stimulation probe (1 cm diameter, Electro Medical Supplies, Wantage, UK) pressed into the femoral triangle and maintained in place by adhesive tape. The surface of the anode, a 10×7 cm carbon rubber electrode (Electro Medical Supplies) was coated in electrode gel and located over the greater trochanter. Square wave pulses, 100 μ s in duration, were delivered via a constant current variable voltage stimulator (Model DS7AH, Digitimer, Ltd, Welwyn Garden City, UK). The precise location of the cathode was determined as the position that evoked the greatest twitch response for a particular submaximal electrical current (typically 30–50 mA). Single impulses were delivered with stepwise increments in the current, separated by 10 s to allow for neuromuscular recovery, until a plateau in the amplitude of twitch force and M-waves were reached. The stimulus intensity was then

increased by 20% above the value required to elicit a plateau to ensure supramaximal stimulation, and three discrete supramaximal stimuli, each separated by 10 s were then delivered. The mean M-wave peak-to-peak amplitude of the three supramaximal stimuli was defined as the maximal M-wave ($M_{\max}$) and used for normalization of voluntary quadriceps EMG. Force measurements from the evoked twitches included Twitch F_{25} and F_{50} , at 25 and 50 ms after force onset, Twitch peak force, Twitch RFD_{0-50} , and these were averaged across the three maximal twitch contractions.

The current was reduced prior to commencing the octet contractions (eight pulses at 300 Hz). Stepwise increments in the current were delivered 15 s apart until the supramaximal current used for twitch contractions was attained. Thereafter, three octet contractions were evoked and recorded with this supramaximal current. Octet contractions of this type have been shown to evoke

the maximum possible RFD from the MTU (Deutekom et al., 2000; de Ruiter et al., 2004). In the present study, real-time inspection of octet force and the slope of the force-time curve confirmed a plateau in both peak octet force and peak RFD during the incremental stimulation. Variables recorded and averaged across the three supramaximal octet contractions included Octet F_{25} , F_{50} , and F_{75} at 25, 50, and 75 ms from force onset, Octet peak force, and Octet RFD_{0-50} . Twitch and Octet forces were also normalized to MVF.

Knee flexion MVCs

Knee flexion strength was assessed at the same joint positions and involved the same procedures and instructions as the knee extension MVCs, but with participants performing flexion of the knee. Strong verbal encouragement and biofeedback were given during all MVCs. Maximal RMS EMG of the BF and LH was calculated over a 500-ms period surrounding peak force (250 ms either side), averaged for the hamstrings group and defined as $HEMG_{max}$.

Data analyses

EMG and force signal onsets (voluntary and evoked) were identified with a standardized systematic protocol of visual identification using a standard method by two trained experienced investigators (Tillin et al., 2010, 2013b; Buckthorpe et al., 2012; Hannah et al., 2012). Visual inspection is considered to provide highly accurate event detection (Walter, 1984; Staude & Wolf, 1999), and this method has been used as a gold standard method in a number of studies (Hodges & Bui, 1996; Staude, 2001; Allison, 2003) and widely used as the detection method of choice (Carey et al., 1983; Hodges et al., 2001; Pulkovski et al., 2008). While automated methods of signal onset detection have been suggested to provide greater objectivity and have been quite widely used in exercise science, in part due to the reduced analysis time required, their reliability and validity have also been questioned (Tillin et al., 2013b). For example, two recent studies found automated methods to provide no benefit in terms of reliability (Soda et al., 2010; Uliam Kuriki et al., 2011).

Force and EMG recordings were initially viewed with consistent y-axis scales of ~ 1.5 N and 0.05 mV, and an x-axis scale of 500 ms. These scales provided sufficient resolution for the accurate discrimination of signal onset, which was defined as the last peak or trough before the signal deflected away from baseline noise. A vertical cursor was then placed on the signal onset before the signals were viewed at a higher resolution (~ 0.75 N and 0.02 mV for force and EMG, and an x-axis of 25 ms) to verify the accurate placement of the cursor. The slope of the force-time trace was also displayed to confirm the instant of force onset (Tillin et al., 2010). Onsets were determined by two trained experienced investigators.

In order to assess intra- and interobserver reliability for determining force and EMG onsets within this study, repeated analysis of the three best explosive voluntary contractions of 15 participants (equating to 45 force onsets and 135 agonist muscle EMG onsets) was performed. The reliability of force onset determination was very high [standard deviation (SD): intraobserver 0.3 ms, interobserver 0.5 ms; mean absolute difference: intraobserver 0.5 ms, interobserver 0.7 ms], and EMG onset determination was also reliable (SD: intraobserver 0.6 ms, interobserver 1.3 ms; mean absolute difference: intraobserver 0.9 ms, interobserver 1.9 ms).

Statistical analyses

Descriptive statistics are presented as mean $\pm$ standard deviation, and CV was also used to describe the between-participant vari-

ability. All explosive force and sequential RFD values were expressed in absolute terms (N and N/s) and relative to MVF (%MVF and MVF/s). Statistical analyses were performed using the SPSS version 16 (SPSS Inc., Chicago, Illinois, USA) and statistical significance was set at $P < 0.05$. Paired t -tests were used to compare volitional and evoked explosive force values.

Preliminary analysis involved calculating Pearson's product moment bivariate correlations between explosive voluntary force and individual predictor variables (QEMG, HEMG, Twitch F, Octet F, MVF). These relationships were considered in absolute and relative terms (i.e., with both predictor and dependent variables expressed in the same way). Bivariate correlations were also calculated between voluntary sequential RFD (successive 50-ms time periods) and predictor variables ($\Delta QEMG/\Delta t$, $\Delta HEMG/\Delta t$, Twitch RFD_{0-50} , Octet RFD_{0-50} , MVF, Prior Force).

More detailed analysis involved stepwise multiple linear regressions, which simultaneously examined the influence of all the entered predictor variables, in order to reveal those that independently explained a significant proportion of the total variance. Predictor variables were typically assessed at the identical time point/period preceding the voluntary force measurement. However, the evoked measures were clearly of shorter duration (Twitch 50 ms; Octet 75 ms) than the voluntary explosive force recordings (150 ms; Fig. 1). The final measurements from the evoked force responses were used as predictor variables for voluntary explosive force production and sequential RFD at later time points/periods.

Results

Interindividual variability

Knee extension MVF was 459 ± 96 N (CV 21%) and ranged from 300 to 693 N. The interindividual variability in explosive force production was greatest during the early phase of contraction (e.g., F_{50} , range 14–182 N or 13-fold, CV 48%), but decreased throughout the contraction [F_{150} , range 215–536 N, CV 23%; Fig. 2(a)]. These differences were only partially reduced by normalization to MVF, as relative explosive force remained highly variable during the initial 75 ms of contraction (e.g., relative F_{50} , range 4–30%MVF or 7.5-fold, CV 38%), but showed a greater decrease in variability as the contraction progressed [relative F_{150} , range 58–88%MVF, CV 8%; Fig. 2(b)].

Octet force exhibited less than half the variability between participants as voluntary explosive force (Octet F_{25-75} , CV 19–23%; Table 1). Normalization of Octet F (to MVF) only marginally reduced variability at 25 ms (CV 21%) but had a more pronounced effect at 50 and 75 ms (CV 11–12%). Twitch force was also less variable than voluntary explosive force (Twitch F_{25} and F_{50} , CV 22–27%), and when expressed relative to MVF, this variability was somewhat reduced (CV 16–21%). The maximal capacity of the MTU to produce force explosively, as indicated by Octet F, was typically far greater than most individuals could achieve voluntarily ($P < 0.001$ at 25–75 ms). Consequently, the volitional explosive force achieved was $16 \pm 7\%$ (F_{25}), $45 \pm 19\%$ (F_{50}), and $74 \pm 18\%$ (F_{75}) of Octet F. Twitch F_{25} was also greater than voluntary F_{25} ($P < 0.001$), but at 50 ms, these variables were similar ($P = 0.68$).

Table 1. Mean and interindividual variability of the intrinsic contractile properties of the quadriceps femoris in response to supramaximal evoked twitch and octet contractions. Data are displayed as mean $\pm$ SD (CV, %) ($n = 40$)

Evoked variable	Time point after force onset (ms)			
	25	50	75	Peak
Twitch F (N)	28 $\pm$ 8 (27)	81 $\pm$ 18 (22)	–	99 $\pm$ 26 (27)
Relative Twitch F (%MVF)	6.2 $\pm$ 1.3 (21)	17.7 $\pm$ 2.9 (16)	–	21.5 $\pm$ 3.9 (16)
Octet F (N)	67 $\pm$ 16 (23)	184 $\pm$ 36 (19)	285 $\pm$ 51 (21)	305 $\pm$ 65 (21)
Relative Octet F (%MVF)	14.8 $\pm$ 3.0 (21)	40.5 $\pm$ 4.8 (12)	54.6 $\pm$ 5.8 (11)	66.7 $\pm$ 7.2 (11)

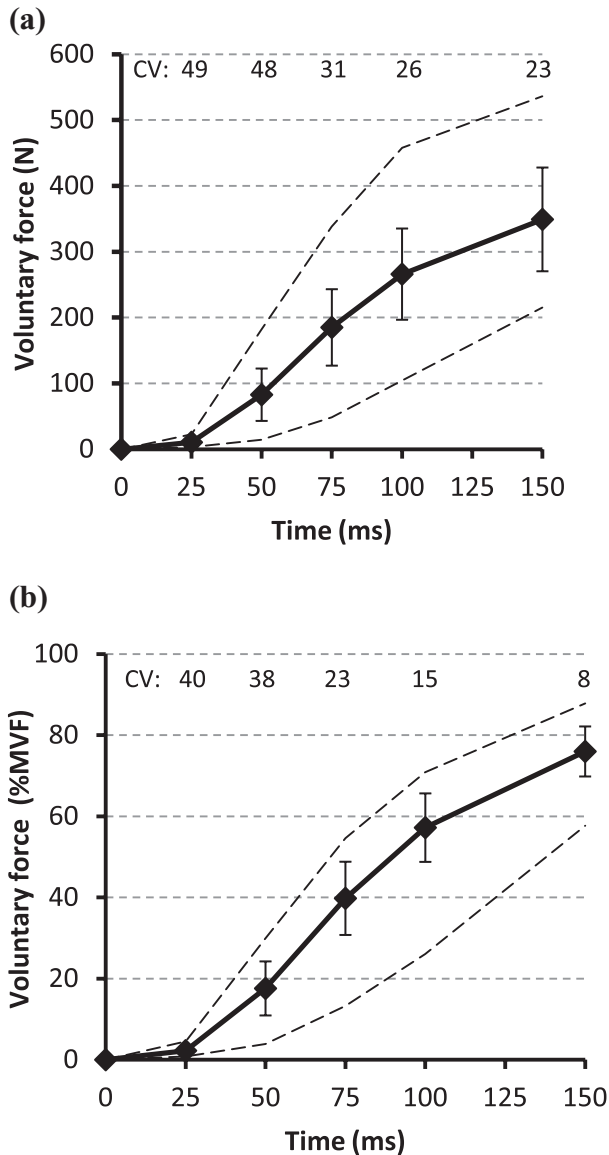


Fig. 2. Interindividual variability in absolute (a) and relative (b) explosive voluntary force production of the knee extensors during the first 150 ms of explosive voluntary contraction. Data are mean $\pm$ SD ($n = 40$; solid line), with maximum and minimum values (dashed lines) and between-subjects coefficient of variation values also shown for each time point.

In comparison to QEMG_{MVF} ($8.8 \pm 2.5\%$ $M_{\max}$), during the explosive voluntary contractions, QEMG was submaximal initially (QEMG_{0–50}, $4.4 \pm 1.7\%$ $M_{\max}$; Fig. 3), but was typically high, or maximal, from 50 ms

onwards (QEMG_{50–100}, 9.3 ± 2.4 and QEMG_{100–150}, 8.9 ± 2.0). In a similar manner to voluntary explosive force, QEMG also showed the most variability during this initial phase compared with later periods (CV: QEMG_{0–50}, 38%; QEMG_{50–100}, 26%; QEMG_{100–150}, 23%). Although antagonist HEMG appeared to be highly variable between participants (CV 71–108%), it was confined to relatively low levels of co-activation (HEMG_{0–50}, 0.7 ± 0.8 ; HEMG_{50–100}, 2.8 ± 2.0 ; HEMG_{100–150}, $3.4 \pm 2.5\%$ EMG_{max}).

Determinants of absolute explosive force and sequential RFD

The multiple regression analysis showed the total variance in explosive force explained by the predictor variables increased throughout the contraction from 59% (F_{25}) to 93% (F_{150}), but the contribution of specific predictor variables changed over the duration of the contraction [Fig. 4(a)]. The total explained variance in sequential RFD declined across the three successive time periods of 0–50, 50–100, and 100–150 ms (77% to 60%) and also showed the influence of specific predictor variables to change with time [Fig. 5(a)].

MVF was correlated increasingly strongly with absolute explosive force as the contraction progressed ($r = 0.59–0.95$, Table 2) and was the primary determinant of the explained variance in the multiple regression analysis from 75 ms onwards [F_{75} , 55%; F_{100} , 75%; F_{150} , 90%; Fig. 4(a)]. MVF was correlated with sequential RFD during each of the three 50-ms time periods ($r = 0.53–0.76$; Table 3), but only contributed modestly to the explained variance in the regression analysis during the third period [RFD_{100–150}, 28%; Fig. 5(a)]. The Prior Force achieved at 100 ms also positively contributed to RFD_{100–150} (32%; i.e., the higher F_{100} , the higher RFD_{100–150}).

Explosive force and RFD during the early phase of contraction were correlated with QEMG (first 50 ms, $r = 0.40–0.61$, Tables 2 and 3). Within the multiple regression analysis, QEMG contributed to the explained variance in explosive voluntary force at all time points, but with a peak contribution at 50 ms [37%; Fig. 4(a)]. Similarly, QEMG only contributed to the variance in RFD_{0–50} (37%) and not during later periods [Fig. 5(a)]. Antagonist HEMG did not correlate with, or contribute to the variance in, absolute explosive force or sequential RFD.

The intrinsic contractile properties, Twitch F ($r = 0.50–0.71$) and Octet F ($r = 0.39–0.85$; Table 2), were, in

the main, increasingly correlated with explosive voluntary F as the contraction progressed up to 150 ms. Multiple regression analysis found Twitch F, rather than Octet F, to make a substantial contribution to the explained variance during the early phase of contraction [F_{25} , 34%; F_{50} , 40%; RFD_{0-50} , 40%; Figs 4(a) and 5(a)]. However, Octet RFD_{0-50} was the primary determinant of voluntary RFD_{50-100} accounting for 68% of the variance during the steepest period of voluntary force development.

Determinants of relative explosive force and sequential RFD

To further highlight the comparative importance of neural and intrinsic contractile characteristics, relative

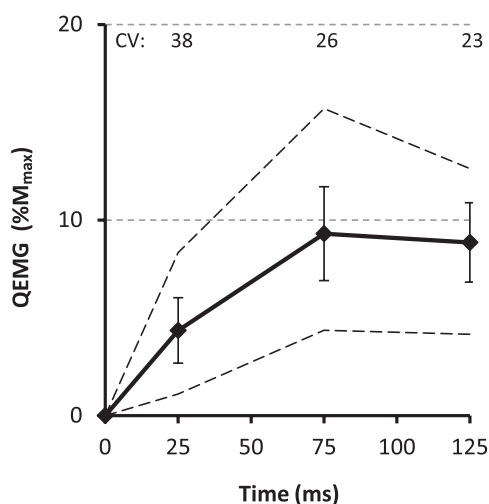


Fig. 3. Interindividual variability in quadriceps EMG during explosive voluntary contractions of the knee extensors. Values are plotted as the central time point for EMG periods 0–50, 50–100 and 100–150 ms. Data are mean $\pm$ SD ($n = 40$; solid line), with maximum and minimum values (dashed lines) and between-subjects coefficient of variation values also shown for each time point.

Table 2. Bivariate correlation coefficients between predictor variables and absolute and relative (%MVF) voluntary explosive force of the knee extensors during the first 150 ms of contraction ($n = 40$). Predictor variables were then considered in the multiple regression analysis (Fig. 4A and B)

Time (ms):	Voluntary explosive force at 25–150 ms									
	Absolute					Relative				
	25	50	75	100	150	25	50	75	100	150
Neural predictor variables										
QEMG	0.40*	0.61**	0.41**	0.21	0.10	0.45**	0.71**	0.68**	0.59**	0.48**
HEMG	–0.21	0.07	0.06	0.07	0.10	–0.21	0.13	0.18	0.29	0.32*
Intrinsic contractile predictor variables:										
Twitch F	0.59**	0.50**	0.54**	0.65**	0.71**	0.25	–0.11	–0.13	–0.11	–0.03
Octet F	0.39*	0.48**	0.64**	0.80**	0.85**	0.00	–0.25	–0.02	0.14	0.27*
Contractile capacity										
MVF	0.59**	0.67**	0.74**	0.87**	0.95**	–	–	–	–	–

* $P < 0.05$; ** $P < 0.01$.

Predictor variables: QEMG and HEMG, quadriceps (% M_{max}) and hamstrings (% EMG_{max}) EMG assessed from onset up to each respective time point; Twitch F and Octet F, evoked twitch and octet force assessed at each respective time point up to a maximum 50 (twitch) or 75 ms (octet), and considered in absolute (vs absolute voluntary force) and relative (vs relative voluntary force) terms; MVF, maximum voluntary force.

values, corrected for MVF, were examined. The total variance in relative explosive force explained by the predictor variables ranged from 35% to 60% during the first 150 ms of contraction (i.e., substantially less than the explained variance in absolute explosive force), with the greatest variance explained at 50 ms [Fig. 4(b)]. The total variance in relative RFD explained by the regression analysis declined from 60% to 31% during the three consecutive 50-ms periods, and the variables contributing to the explained variance also changed over time [Fig. 5(b)].

QEMG was more highly correlated with relative, than absolute, explosive force production at all time points (Table 2). Multiple regression revealed QEMG as the largest determinant of relative voluntary force at all time points [21–51%, Fig. 4(b)] making the greatest contribution to relative F_{50} , although it only contributed to the variance in RFD_{0-50} (37%) and not the later time periods. In contrast, the intrinsic contractile properties (Twitch F and Octet F) were not generally correlated with relative explosive force ($r \leq 0.27$), whereas they were with absolute explosive force ($r \geq 0.39$), and only explained a modest proportion of the total variance in relative explosive force at any time point ($\leq 15\%$ at F_{25} , F_{50} , and F_{150}). However, relative Octet RFD_{0-50} was the only determinant of relative RFD_{50-100} [35%, Fig. 5(b)]. The Prior Force achieved after 100 ms, when expressed relative to MVF, was negatively correlated with relative $RFD_{100-150}$ ($r = -0.56$) explaining 31% of the total variance, which was in contrast to its positive influence on absolute $RFD_{100-150}$.

Discussion

Human explosive force production showed pronounced variability, particularly during the early phase of isometric contraction (F_{50} , 13-fold range; relative F_{50} , 7.5-fold range), being much more variable than MVF (2.3-fold

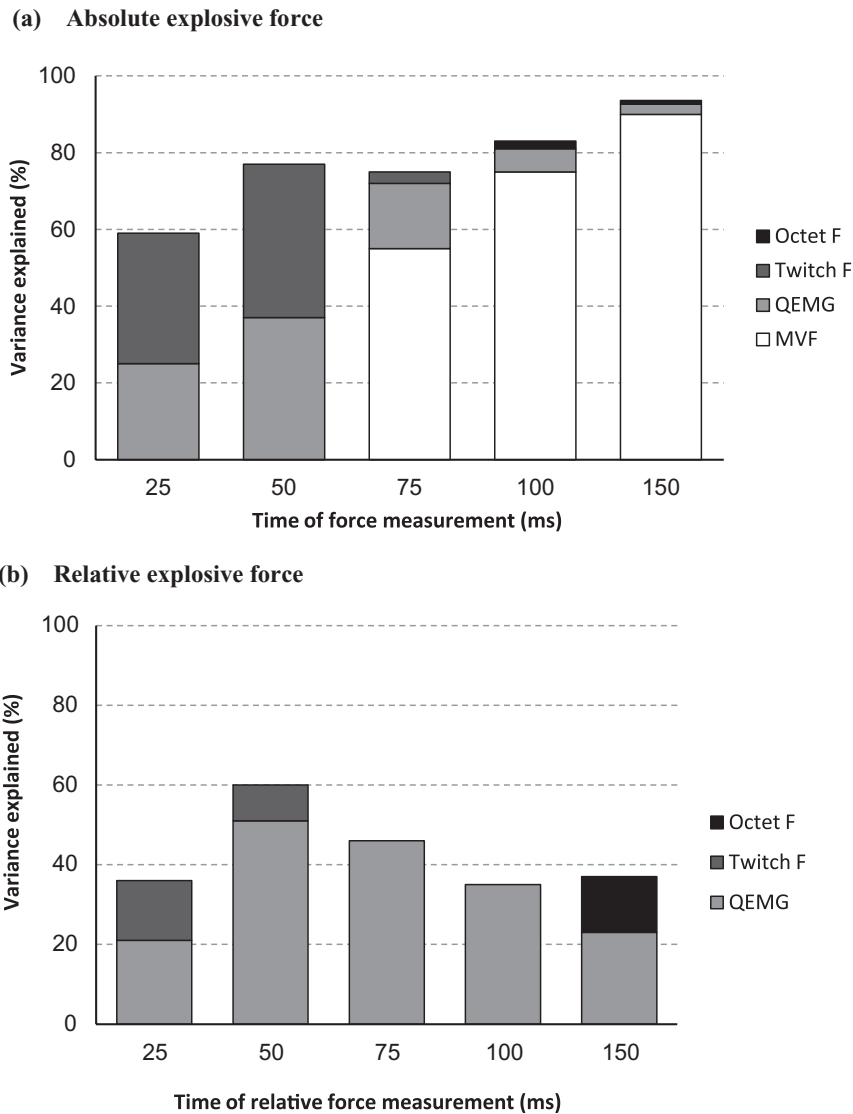


Fig. 4. Determinants of (a) absolute and (b) relative (to MVF) voluntary explosive force of the knee extensors during the first 150 ms of contraction. Predictor variables that independently explained a significant proportion of the total variance assessed with stepwise multiple linear regressions are shown. Predictor variables: Octet F and Twitch F, evoked twitch/octet force assessed at each respective time point up to a maximum 50/75 ms, and considered in absolute (vs absolute voluntary force) and relative (vs relative voluntary force) terms; QEMG, quadriceps EMG (% $M_{\max}$) assessed from onset to each respective time point; MVF, maximum voluntary force ($n = 40$).

range). This observation may have important implications given the apparent functional importance of explosive force production for situations where there is a requirement to produce force quickly (e.g., stabilizing joints to prevent injury, maintaining/correcting balance to prevent a fall, and explosive athletic events). The integrated influence of the predictor variables, within the multiple regressions, explained 59–93% and 35–60% of the variance in absolute and relative explosive force production, respectively. A second major finding was that the primary determinants of explosive force production changed throughout the time course of the contraction. Agonist neuromuscular activation (QEMG) and surprisingly Twitch F, rather than Octet F, were important determinants in the initial 50 ms.

However, Octet RFD_{0–50} was the primary determinant of the steepest phase of voluntary force development (RFD_{50–100}), explaining 68% of the variance once agonist activation was already high/maximal from 50 ms onwards. MVF had an increasing influence on explosive force in the later stages of the contraction (75–150 ms), accounting for 55–90% of the variance. Understanding the role of different determinants has consequences for interventions to improve explosive force production (e.g., exercise training). The present data suggest that neuromuscular activation is a target adaptation for improving early phase, whereas enhancing the contractile properties and particularly MVF may have more relevance for developing late-phase, explosive force production.

Table 3. Bivariate correlation coefficients between predictor variables and absolute and relative (%MVF) voluntary explosive RFD of the knee extensors during three sequential 50 ms time periods from the onset of contraction ($n = 40$). Predictor variables were then considered in the multiple regression analysis [Fig. 5(a) and (b)]

Time period (ms)	Voluntary sequential rate of force development					
	Absolute			Relative		
	0–50	50–100	100–150	0–50	50–100	100–150
Neural predictor variables						
$\Delta QEMG/\Delta t$	0.61**	–0.11	0.31	0.71**	0.37*	0.16
$\Delta HEMG/\Delta t$	0.07	0.06	0.20	0.13	0.25	0.18
Intrinsic contractile predictor variables						
Twitch RFD _{0–50}	0.50**	0.57**	0.40*	–0.11	0.04	0.15
Octet RFD _{0–50}	0.48**	0.83**	0.41*	–0.25	0.59**	0.09
Contractile capacity and history						
MVF	0.67**	0.76**	0.53**	–	–	–
Prior Force	–	0.37*	0.18	–	–0.23	–0.56**

* $P < 0.05$; ** $P < 0.01$.

Predictor variables: $\Delta QEMG/\Delta t$ and $\Delta HEMG/\Delta t$, rate of change in quadriceps (% M_{max}) and hamstrings (% EMG_{max}) EMG compared with the previous time period; Twitch RFD_{0–50} and Octet RFD_{0–50}, evoked twitch and octet rate of force development from 0–50 ms were considered in absolute (vs absolute voluntary RFD) and relative (vs relative voluntary RFD) terms; MVF, maximum voluntary force; Prior Force, force measured at the start of the respective period was also considered in absolute (vs absolute voluntary RFD) and relative (vs relative voluntary RFD) terms.

Interindividual variability

This is the first study to carefully document the wide interindividual variability in explosive force production that was observed despite recruiting a relatively homogeneous cohort in terms of their physical activity (low to moderately active and nonstrength/power trained). The substantial early-phase variability in explosive force production (13-fold range) was only somewhat attenuated when expressed relative to MVF (7.5-fold range), whereas the later phase variability was more markedly reduced by normalization. The current study involved males and females, which likely provided a greater variability in absolute explosive force. However, there were no differences in relative explosive force between males and females (data not shown), indicating that sex differences did not contribute substantially to this variability. We believe the rigidity of the dynamometer, low-noise and high-frequency response strain gauge, and accurate determination of signal onset contributed to the sensitivity of the explosive force measurements. For example, in pilot work using video digitization and electrogoniometry, we have found knee joint movement of $\leq 4^\circ$ from rest to MVF with this dynamometer, compared with $> 15^\circ$ with some commercial dynamometers (Tsaopoulos et al., 2007).

This extensive variability likely has important implications for situations where there is a requirement for force to be produced quickly. This includes success in explosive athletic events (Kuitunen et al., 2002; Tillin et al., 2013a) as well as postural corrections during ambulation or following a disturbance of balance that could lead to a fall or injury (Fleming et al., 1991; Izquierdo et al., 1999; Chang et al., 2005; Pijnappels

et al., 2008). For example, noncontact anterior cruciate ligament injuries appear to occur very soon after foot contact with the ground (≤ 50 ms; Krosshaug et al., 2007), and thus, the variability in explosive force production could be an important risk factor. However, musculo-skeletal stabilization in response to perturbed balance may involve different neural processes (sensory feedback and reflex responses), compared with the voluntary actions documented in the current study. While some preliminary research has considered the importance of quick neuromuscular activation in injury-related situations (Lofvenberg et al., 1995; Konradsen et al., 1998; Bloem et al., 2000; Zebis et al., 2009) and athletic tasks (Pääsuke et al., 2001; Tillin et al., 2013a), these have typically involved relatively small groups that may not reflect the full diversity in the general population. Athletic tasks and postural corrections also involve explosive dynamic contractions that have received very little research attention, but appear to involve different activation strategies and capabilities than isometric contractions (Tillin et al., 2012b). Moreover, further research on the implications of the wide variability in explosive force production that we have documented is required.

The simple observation that the interindividual variability of explosive voluntary force was more than 2–3-fold higher than that of the evoked octet force (absolute values, CV: Octet F_{50} 19% vs Voluntary F_{50} 48%; relative values, CV: Octet F_{50} 12% vs Voluntary F_{50} 38%) indicates that volitional neural drive contributes substantially to the between-subject variance during the early phase of voluntary explosive effort. Furthermore, in the present study, agonist neuromuscular activation during this early phase of contraction also showed high interindividual

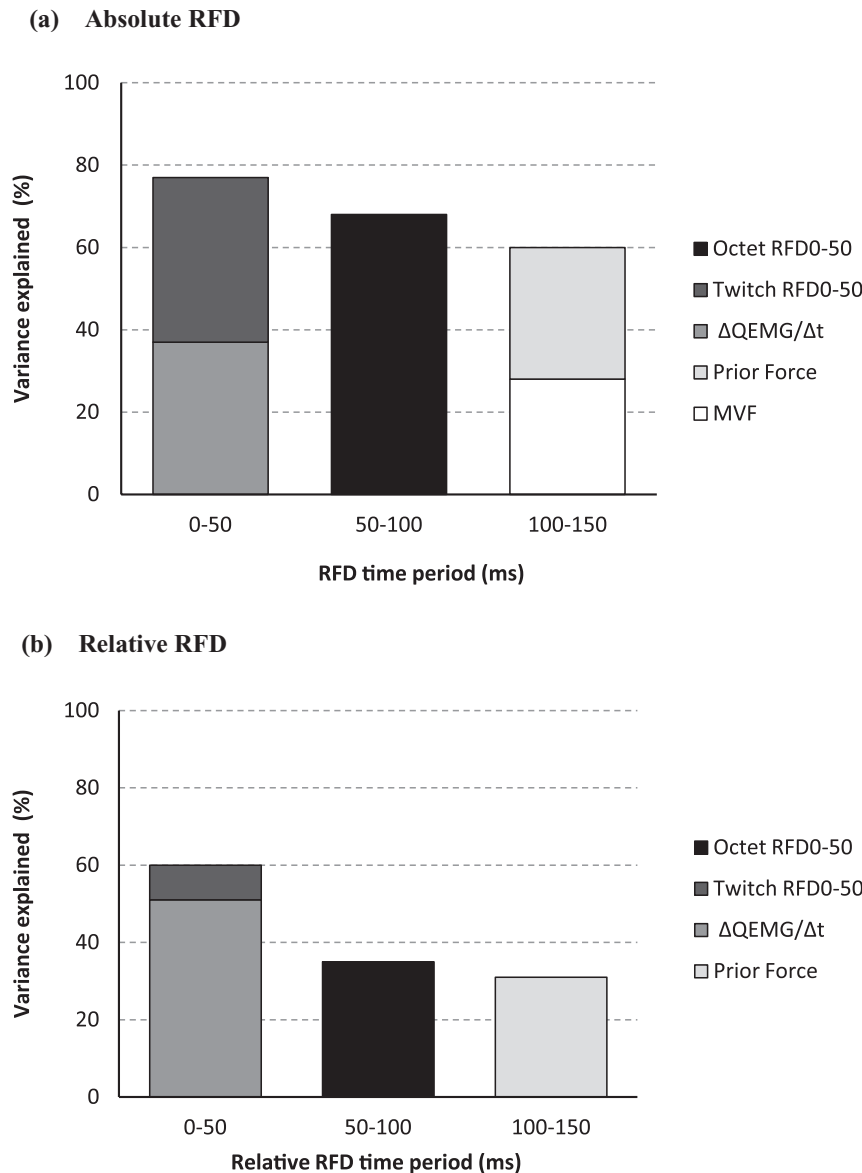


Fig. 5. Determinants of (a) absolute and (b) relative (to MVF) voluntary explosive RFD of the knee extensors during three sequential 50 ms time periods. Predictor variables that independently explained a significant proportion of the total variance assessed with stepwise multiple linear regressions are shown. Predictor variables: Octet RFD₀₋₅₀ and Twitch RFD₀₋₅₀, evoked rate of force development from 0–50 ms, and considered in absolute (vs absolute RFD) and relative (vs relative RFD) terms; $\Delta QEMG/\Delta t$, rate of change in quadriceps EMG compared with the preceding period; Prior Force, force measured at the start of the respective period considered in absolute (vs absolute RFD) and relative (vs relative RFD) terms; MVF, maximum voluntary force; ($n = 40$).

variability (CV: QEMG₀₋₅₀ 38%) that was comparable to that of F₅₀.

Neural determinants of explosive force

Agonist QEMG contributed to the explained variance in absolute explosive force throughout the entire 150 ms of rising muscle force, but particularly during the initial phase (F₂₅₋₇₅, 17–37%). QEMG explained a greater proportion of the variance in relative explosive force (relative F₂₅₋₇₅, 21–51%) and was the strongest determinant at all time points. Therefore, these findings demonstrate the importance of agonist activation for early-phase explosive

sive force production, especially an individual's ability to express their available force generating capacity. Previous findings with small cohorts have also reported strong correlations between agonist EMG and measures of early-phase explosive voluntary force [impulse 0–40 ms (de Ruiter et al., 2006); relative RTD 0–80 ms (Klass et al., 2008)]. Additionally, less direct evidence also supports the notion that neural drive is important for early-phase explosive force production. For example, the greater explosive force capability of explosive power athletes has in part been attributed to differences in neural drive assessed by surface EMG (Tillin et al., 2010), and strength training studies have found enhanced

explosive force production/RFD post-training to be associated with increased EMG amplitude (Van Cutsem et al., 1998; Aagaard et al., 2002; Tillin et al., 2012a). Considering that the participants in the current study had a relatively low physical activity status, the large variability in agonist activation (CV: QEMG₀₋₅₀ 38%) could be largely innate.

Although antagonist HEMG was highly variable (CV, 71–108%), the magnitude of co-activation values was relatively low (<10% HEMG_{max}), and it did not contribute to the variability in explosive force. While we did not measure neuromuscular activation of stabilizer muscles (e.g., hip joint and pelvic girdle stabilizers), given the rapid rise in force with an evoked octet that presumably involves minimal stabilizer activation, the role of stabilizer activation in this restrained isometric situation appears limited. On the other hand, in the voluntary situation, limited stabilizer activation could be a constraint to agonist neural drive, and future work should address the importance of intermuscular activation.

Intrinsic contractile determinants of explosive force

Twitch force, rather than octet force, was the more important determinant of explosive force during the initial 50 ms of contraction explaining 34–40% of the total variance. The greater commonality between the early phase of volitional contraction and a twitch response could reflect a submaximal firing frequency and/or incomplete Ca²⁺ saturation during the initial phase of volitional contraction. A previous study also highlighted the relationship between Twitch F and early-phase voluntary RFD based on correlation analysis (Andersen & Aagaard, 2006), finding similar *r* values to the current study.

The response to the evoked octet was, however, the primary determinant of the steepest phase of voluntary force development (68%, RFD₅₀₋₁₀₀). Therefore, it appears that once agonist neural activation was typically high/maximal after the first 50 ms (based on QEMG), then the subsequent rise in force from 50–100 ms was highly dependent on the contractile properties of the MTU. Previous reports have indicated an influence of fiber type composition (Viitasalo & Komi, 1978) and MTU stiffness (Bojsen-Møller et al., 2005) on voluntary explosive force, and these morphological factors could explain the relationship between evoked octet and voluntary RFD₅₀₋₁₀₀. However, further research is required to more precisely unravel the specific MTU characteristics that are important for voluntary and evoked explosive force production.

The moderate to high bivariate correlations between twitch and octet force with voluntary explosive F₅₀₋₁₅₀ (*r* = 0.48–0.84), were, on the whole, no longer present when MVF was factored out (i.e., relative values). Relative explosive force production, with MVF factored out, was expected to highlight the role of neural vs contractile

determinants. It might have been expected that the intrinsic contractile properties (evoked responses) would have had a stronger influence on relative than absolute explosive force production. However, the fact that this was not the case (the relative evoked responses were unrelated to relative explosive force) indicates a shared variance of the absolute twitch and octet force with MVF. Furthermore, this suggests that the intrinsic contractile properties are a relatively minor determinant of relative explosive force, which was reflected by their minor contribution to the total variance in relative explosive force within the multiple regression analysis (≤15%).

Contractile capacity and history

The influence of MVF on explosive force increased throughout the contraction, accounting for the majority of the variation in the later phase of rising muscle force (100 ms, 75%; 150 ms, 90%), which is similar to the relationship documented by a previous report [52–81% of the variance from 100 to 200 ms (Andersen & Aagaard, 2006)]. As MVF defines the ultimate ceiling/plateau of voluntary force production, it seems logical that as an explosive contraction progresses toward this ceiling (>50% MVF, typically after ~90 ms), explosive force would become increasingly influenced by MVF. Considering the contractile history or Prior Force achieved by 100 ms, this variable contributed to the explained variance in absolute RFD₁₀₀₋₁₅₀ (32%) in a positive manner, but when expressed in relative terms was a negative determinant of relative RFD₁₀₀₋₁₅₀ (31%). While this might appear contradictory, a greater absolute F₁₀₀ is likely to be related to continued high rates of force development, whereas a high relative F₁₀₀ may reduce the scope for relative RFD thereafter.

In conclusion, the pronounced variability in human explosive force production was explained by a range of predictor variables that changed throughout the contraction. Agonist neural activation (QEMG) and surprisingly Twitch F, rather than Octet F, were important determinants in the initial 50 ms. From 50 ms onwards, when agonist activation was typically maximal, the contractile capacity of the MTU for RFD, specifically Octet RFD₀₋₅₀, became the most important determinant of RFD during the second 50-ms period. Explosive force achieved at ≥75 ms was highly determined by MVF and the continued rise in relative force from 100 to 150 ms was negatively influenced by the %MVF already achieved at 100 ms.

Perspectives

This study determined the interindividual variability in explosive force production of the knee extensors, and assessed the neural and contractile determinants of explosive force production. Explosive force production exhibited substantial interindividual variability, particularly

during the early phase of contraction, which may have important consequences for functional situations where there is a need to produce force quickly, e.g., sprinting, jumping, and stabilization following a loss of postural balance. This variability was largely explained by the integrated contribution of predictor neural and contractile variables, but the relative contribution of the most impor-

tant determinants changed during the phase of contraction, and this finding may be useful for designing interventions (e.g., exercise training) to improve explosive force production.

Key words: explosive strength, neural drive, contractile properties.

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